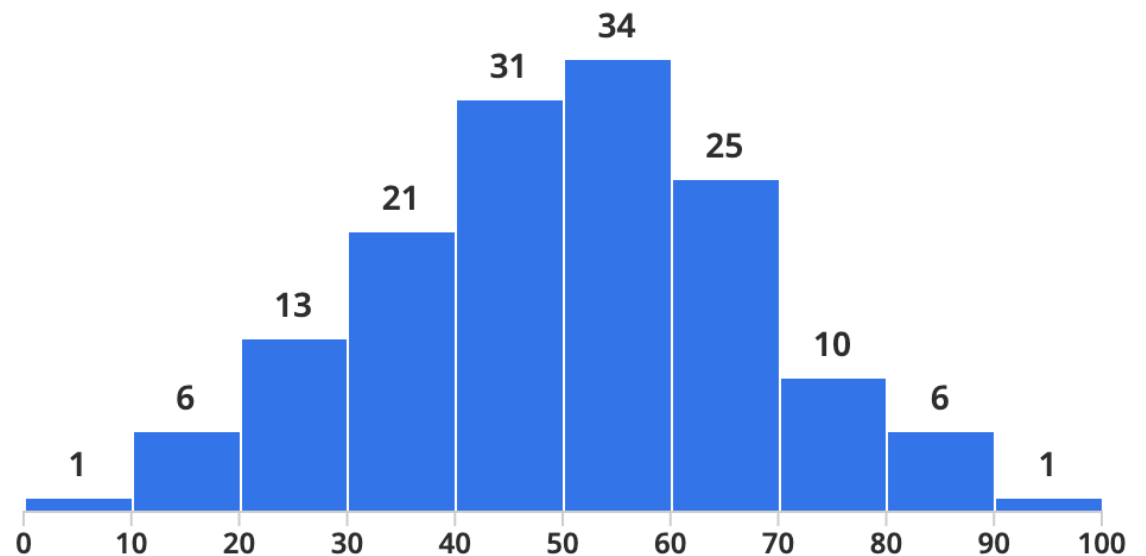


Midterm Grades

- Median: 50
- Mean: 49.55
- Max: 90
- Min: 7

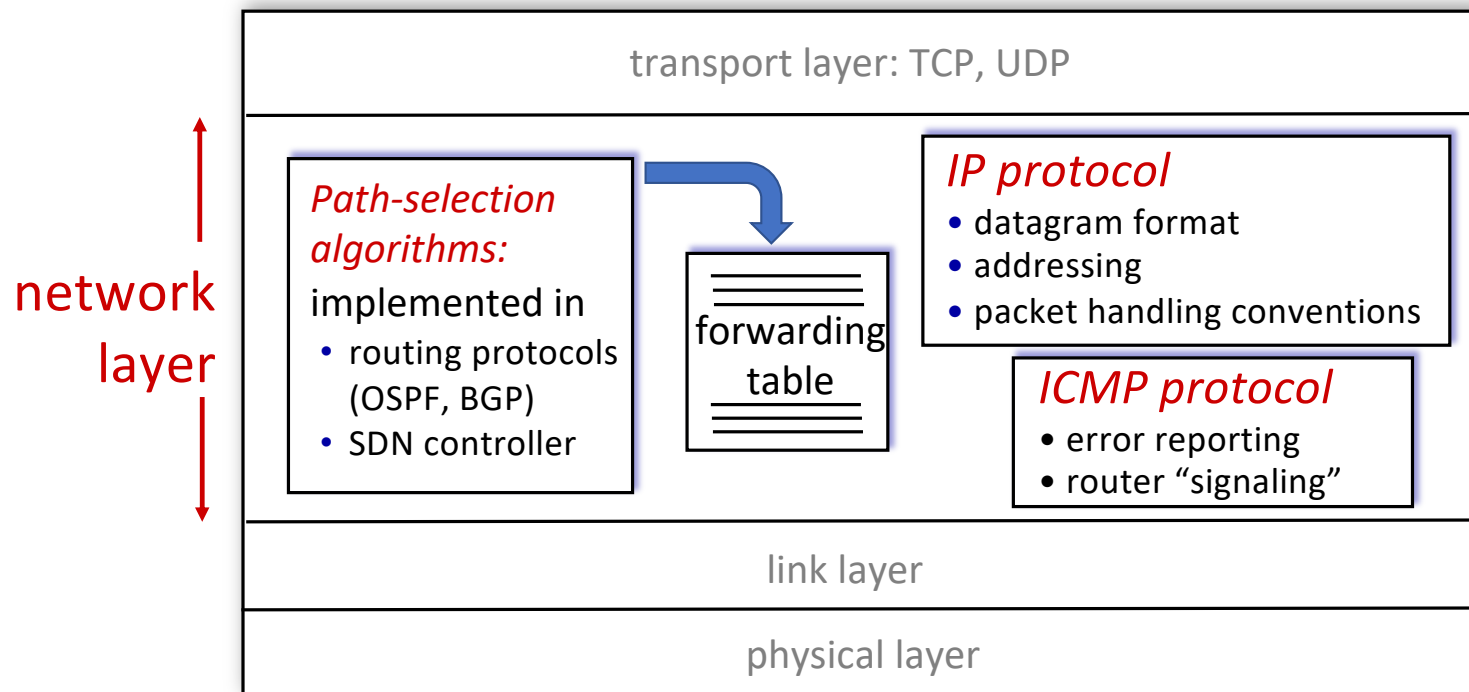


Network layer: data plane roadmap

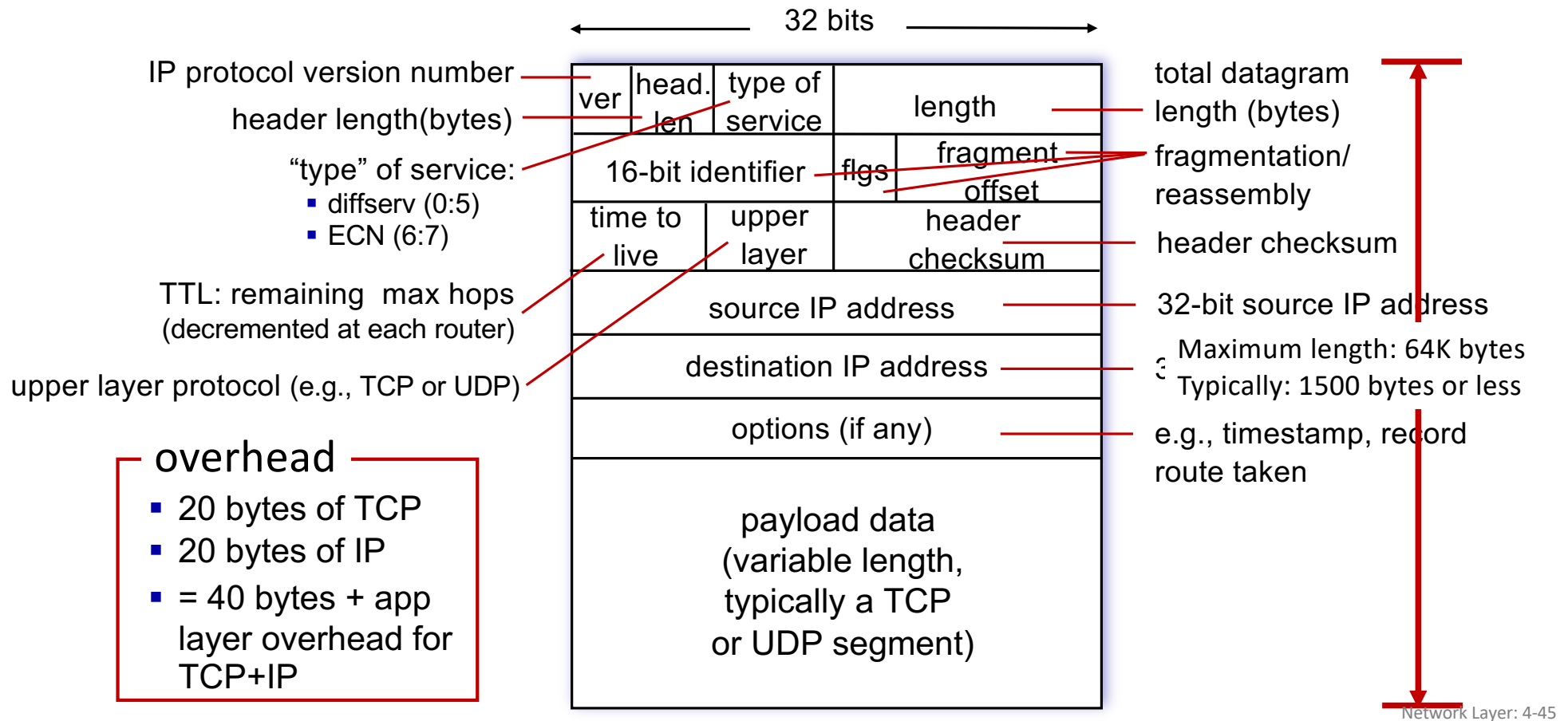
- Network layer: overview
 - data plane
 - control plane
- What's inside a router
 - input ports, switching, output ports, buffer management, scheduling
- IP: the Internet Protocol
 - datagram format, addressing, network address translation, IPv6
- Generalized Forwarding, SDN
- Middleboxes

Network Layer: Internet

host, router network layer functions:

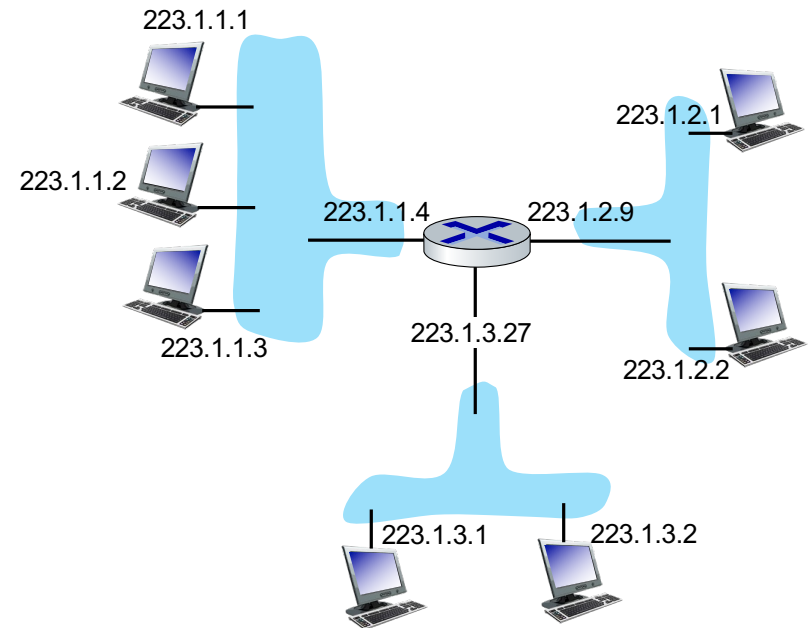


IP Datagram format



IP addressing: introduction

- **IP address:** 32-bit identifier associated with each host or router *interface*
- **interface:** connection between host/router and physical link
 - router's typically have multiple interfaces
 - host typically has one or two interfaces (e.g., wired Ethernet, wireless 802.11)



dotted-decimal IP address notation:

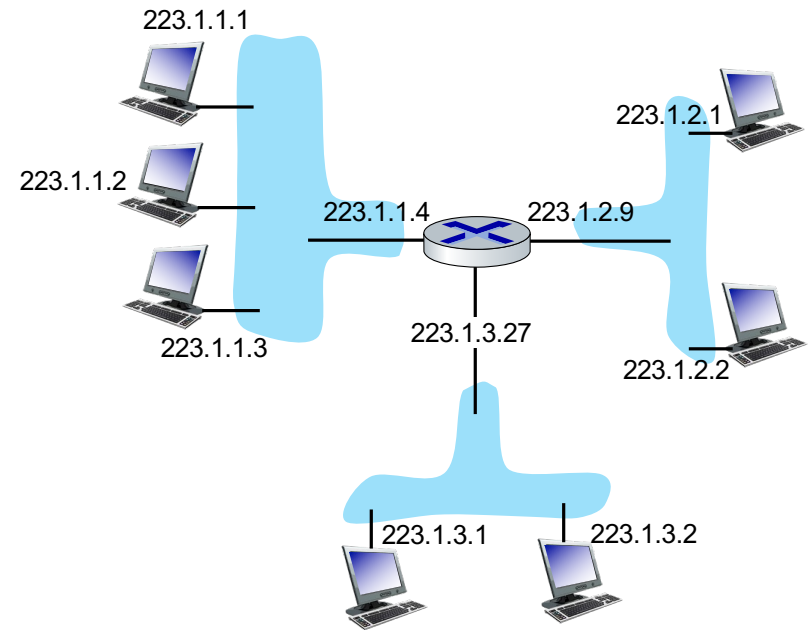
223.1.1.1 = 11011111 00000001 00000001 00000001

223 1 1 1

Network Layer: 4-46

IP addressing: introduction

- **IP address:** 32-bit identifier associated with each host or router *interface*
- **interface:** connection between host/router and physical link
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dotted-decimal IP address notation:

223.1.1.1 = 11011111 00000001 00000001 00000001

223 1 1 1

Network Layer: 4-47

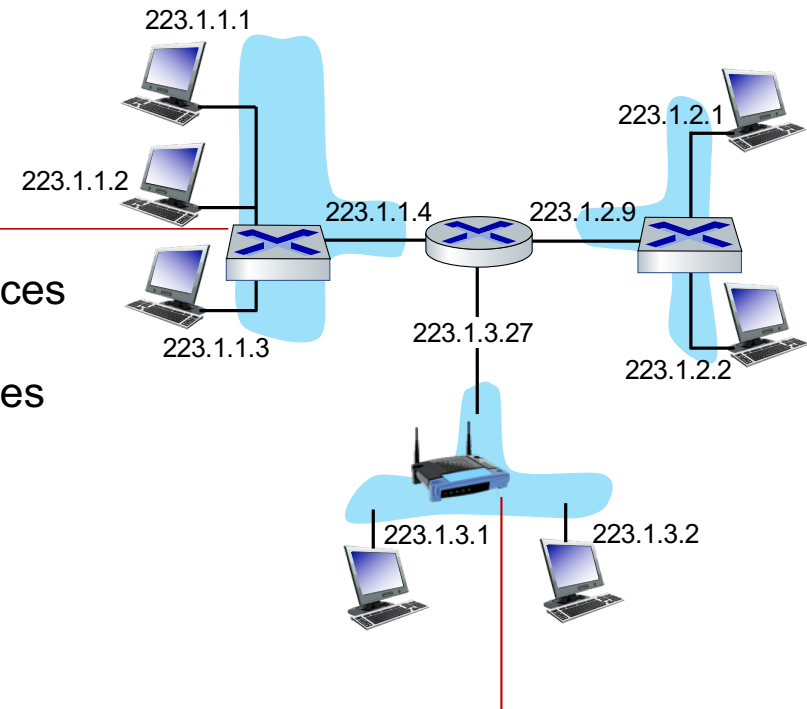
IP addressing: introduction

Q: how are interfaces
actually connected?

A: we'll learn about
that in chapters 6, 7

For now: don't need to worry
about how one interface is
connected to another (with no
intervening router)

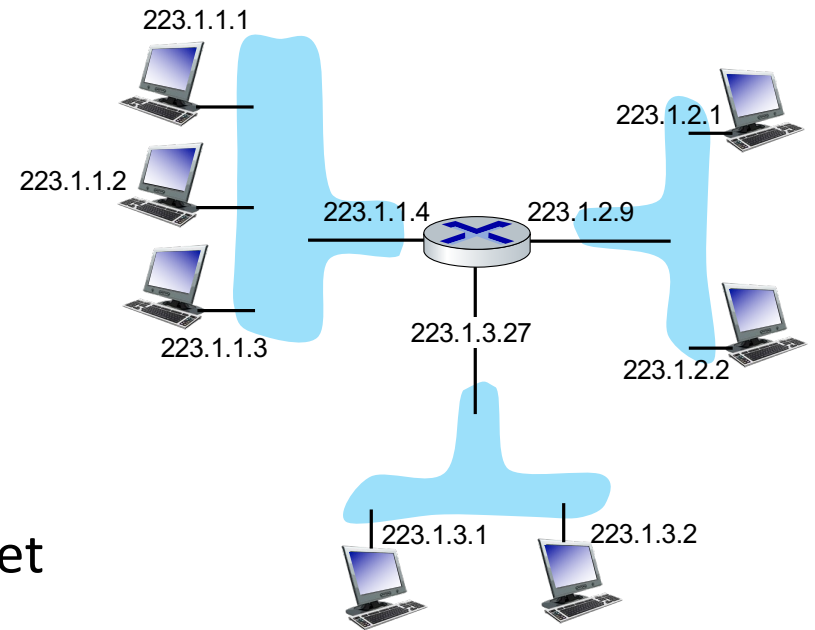
A: wired
Ethernet interfaces
connected by
Ethernet switches



A: wireless WiFi interfaces
connected by WiFi base station

Subnets

- *What's a subnet ?*
 - device interfaces that can physically reach each other **without passing through an intervening router**
- IP addresses have structure:
 - **subnet part**: devices in same subnet have common high order bits
 - **host part**: **remaining** low order bits

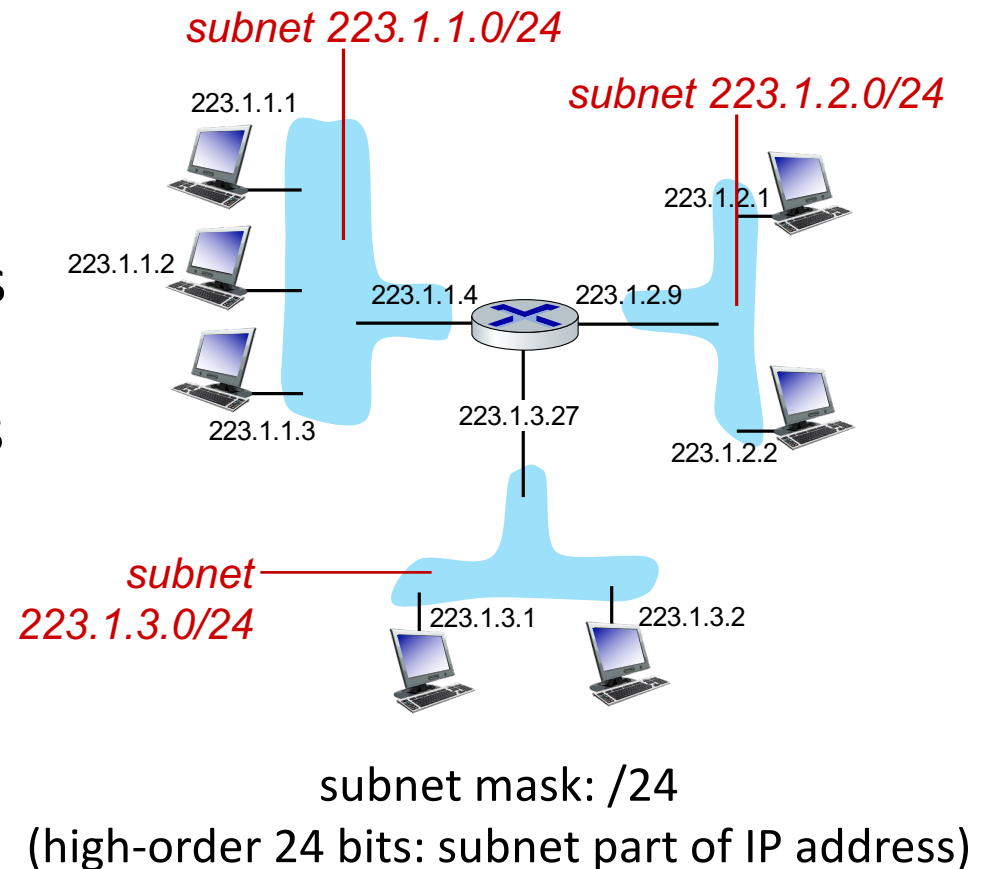


network consisting of 3 subnets

Subnets

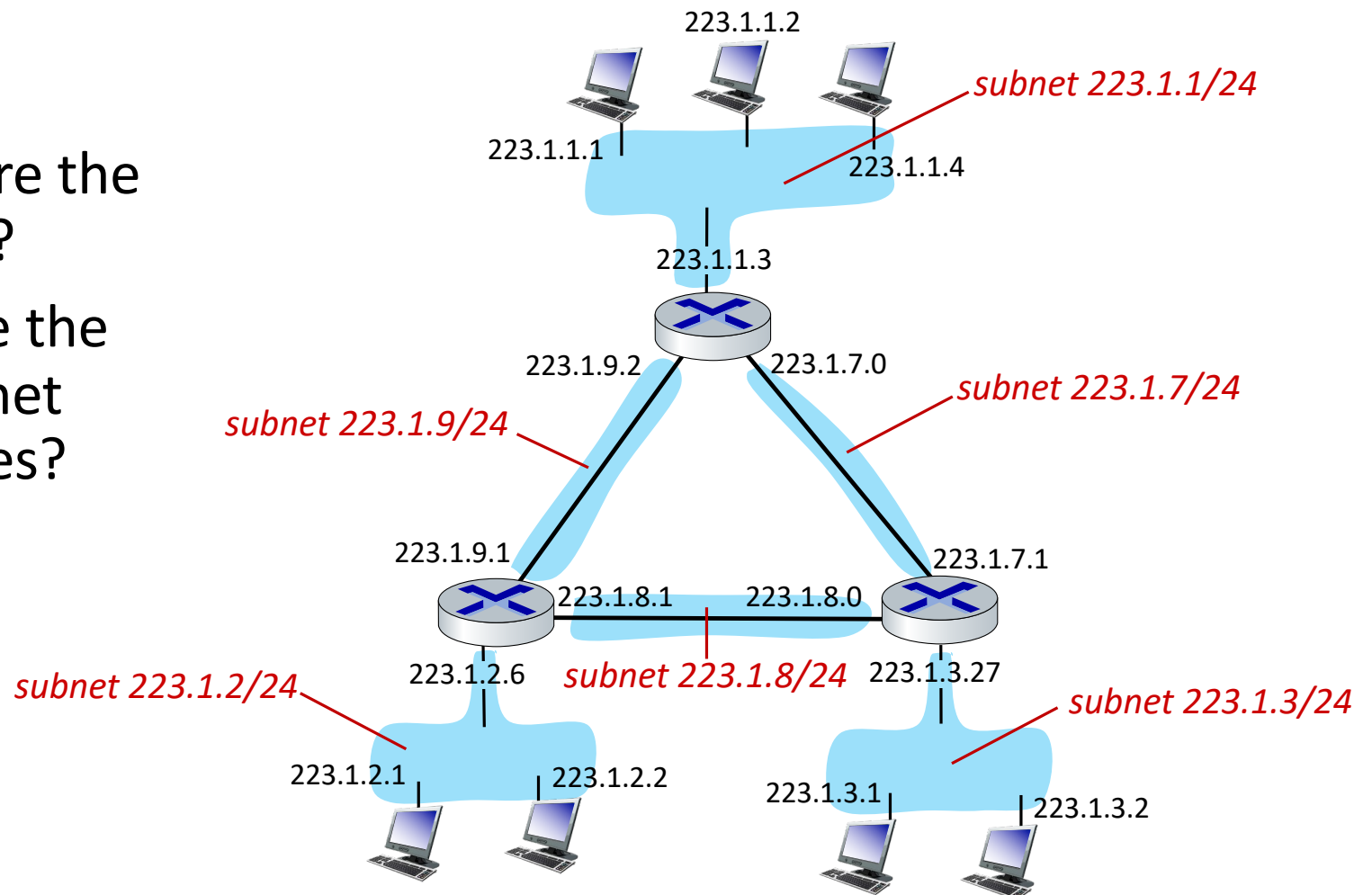
Recipe for defining subnets:

- detach each interface from its host or router, creating “islands” of isolated networks
- each isolated network is called a *subnet*



Subnets

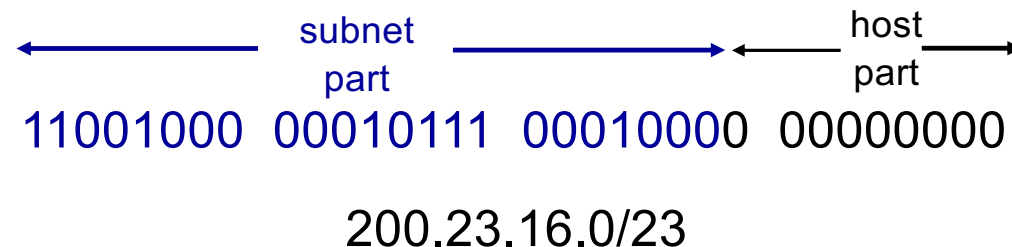
- where are the subnets?
- what are the /24 subnet addresses?



IP addressing: CIDR

CIDR: Classless **I**nter**D**omain **R**outing (pronounced “cider”)

- subnet portion of address of arbitrary length
- address format: **a.b.c.d/x**, where x is # bits in subnet portion of address



IP addresses: how to get one?

That's actually **two** questions:

1. Q: How does a *host* get IP address within its network (host part of address)?
2. Q: How does a *network* get IP address for itself (network part of address)

How does *host* get IP address?

- hard-coded by sysadmin in config file (e.g., /etc/rc.config in UNIX)
- **DHCP**: **D**ynamic **H**ost **C**onfiguration **P**rotocol: dynamically get address from a server
 - “plug-and-play”

DHCP: Dynamic Host Configuration Protocol

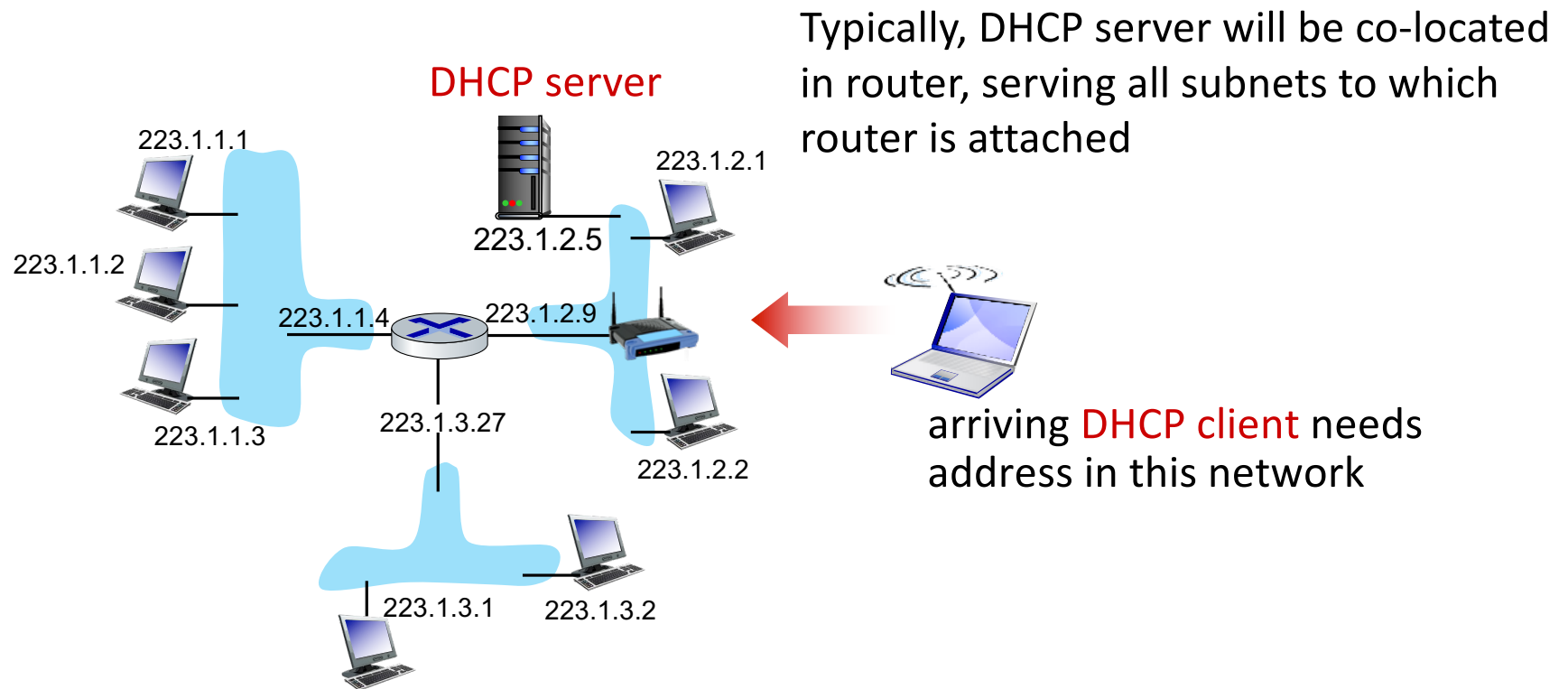
goal: host *dynamically* obtains IP address from network server when it “joins” network

- can renew its lease on address in use
- allows reuse of addresses (only hold address while connected/on)
- support for mobile users who join/leave network

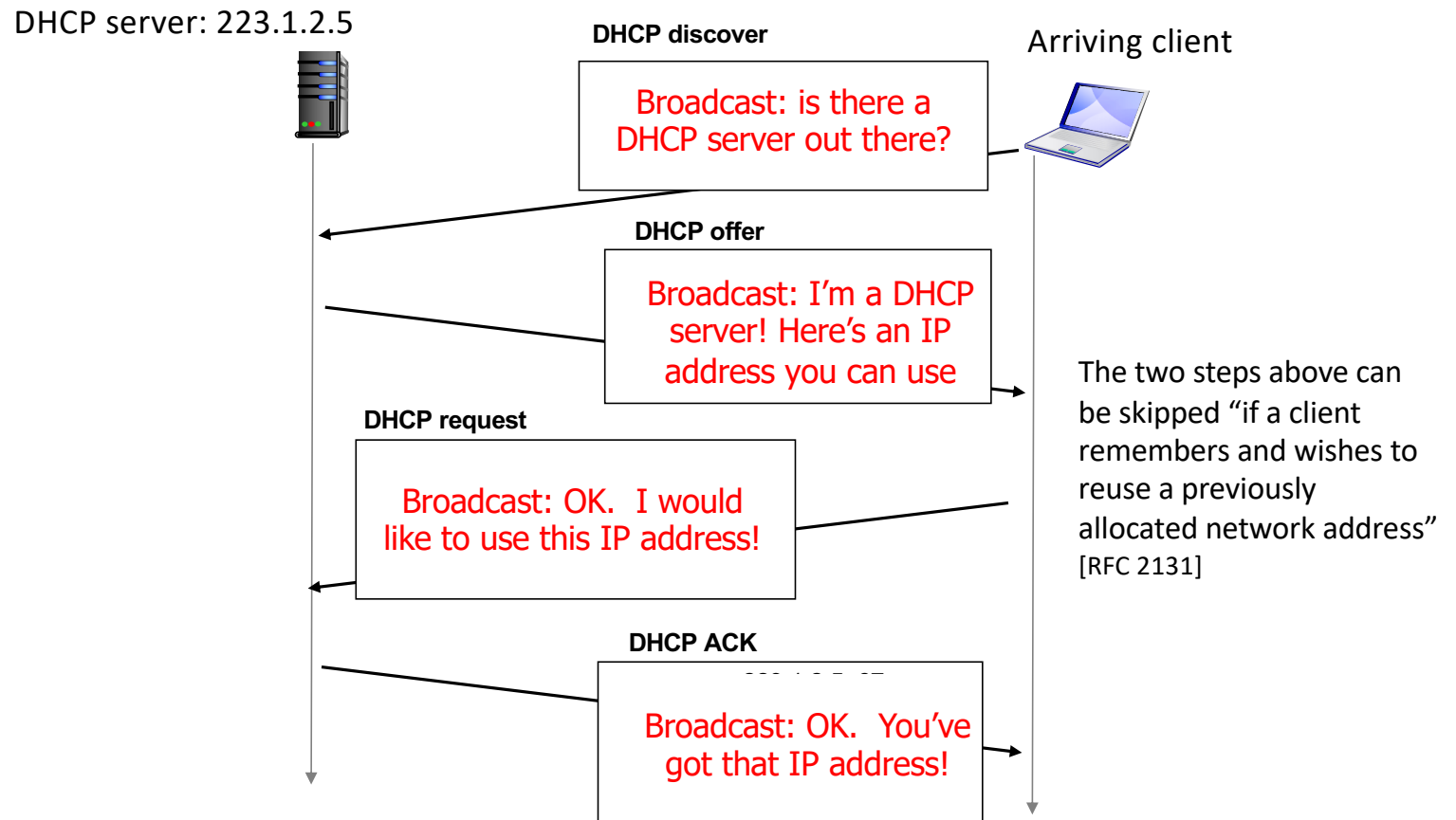
DHCP overview:

- host broadcasts **DHCP discover** msg [optional]
- DHCP server responds with **DHCP offer** msg [optional]
- host requests IP address: **DHCP request** msg
- DHCP server sends address: **DHCP ack** msg

DHCP client-server scenario



DHCP client-server scenario

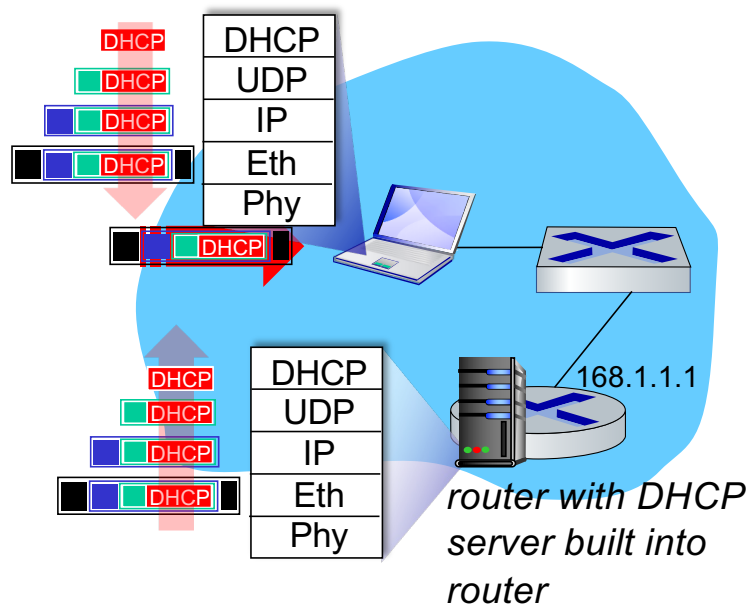


DHCP: more than IP addresses

DHCP can return more than just allocated IP address on subnet:

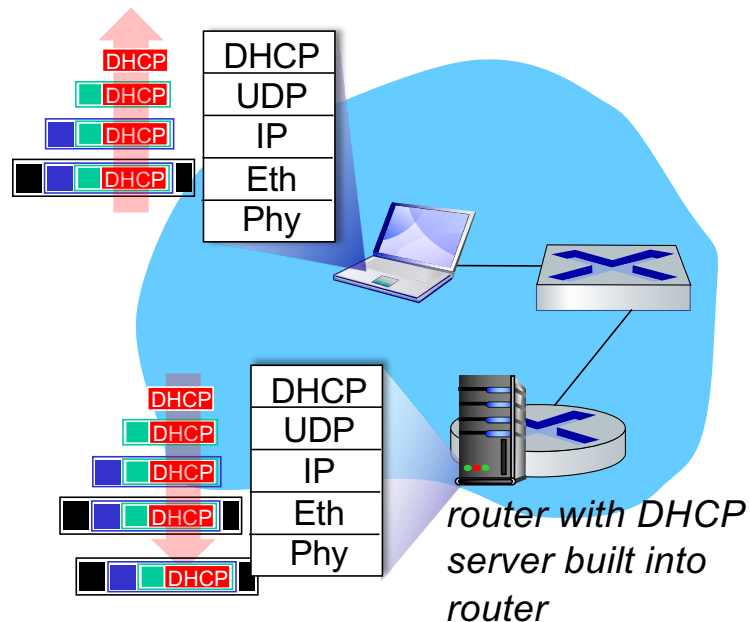
- address of first-hop router for client
- name and IP address of DNS sever
- network mask (indicating network versus host portion of address)

DHCP: example



- Connecting laptop will use DHCP to get IP address, address of first-hop router, address of DNS server.
- DHCP REQUEST message encapsulated in UDP, encapsulated in IP, encapsulated in Ethernet
- Ethernet frame broadcast (dest: FFFFFFFFFFFFFFFF) on LAN, received at router running DHCP server
- Ethernet de-mux'ed to IP de-mux'ed, UDP de-mux'ed to DHCP

DHCP: example



- DHCP server formulates DHCP ACK containing client's IP address, IP address of first-hop router for client, name & IP address of DNS server
- encapsulated DHCP server reply forwarded to client, de-muxing up to DHCP at client
- client now knows its IP address, name and IP address of DNS server, IP address of its first-hop router

IP addresses: how to get one?

Q: how does *network* get subnet part of IP address?

A: gets allocated portion of its provider ISP's address space

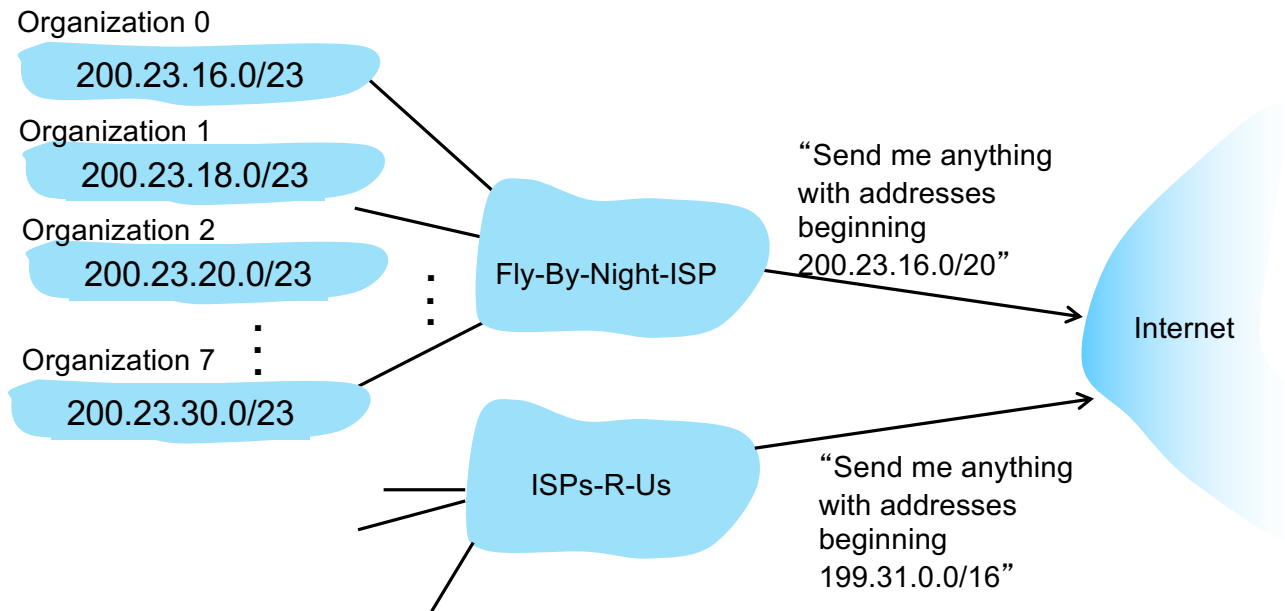
ISP's block 11001000 00010111 00010000 00000000 200.23.16.0/20

If ISP allocates out its address space in 8 blocks:

Organization 0	<u>11001000</u>	<u>00010111</u>	<u>00010000</u>	00000000	200.23.16.0/23
Organization 1	<u>11001000</u>	<u>00010111</u>	<u>00010010</u>	00000000	200.23.18.0/23
Organization 2	<u>11001000</u>	<u>00010111</u>	<u>00010100</u>	00000000	200.23.20.0/23
...					
Organization 7	<u>11001000</u>	<u>00010111</u>	<u>00011110</u>	00000000	200.23.30.0/23

Hierarchical addressing: route aggregation

hierarchical addressing allows efficient advertisement of routing information:



IP addressing: last words ...

Q: how does an ISP get block of addresses?

A: ICANN: Internet Corporation for Assigned Names and Numbers
<http://www.icann.org/>

- allocates IP addresses, through 5 regional registries (RRs) (who may then allocate to local registries)
- manages DNS root zone, including delegation of individual TLD (.com, .edu , ...) management

Q: are there enough 32-bit IP addresses?

- ICANN allocated last chunk of IPv4 addresses to RRs in 2011
- NAT (next) helps IPv4 address space exhaustion
- IPv6 has 128-bit address space

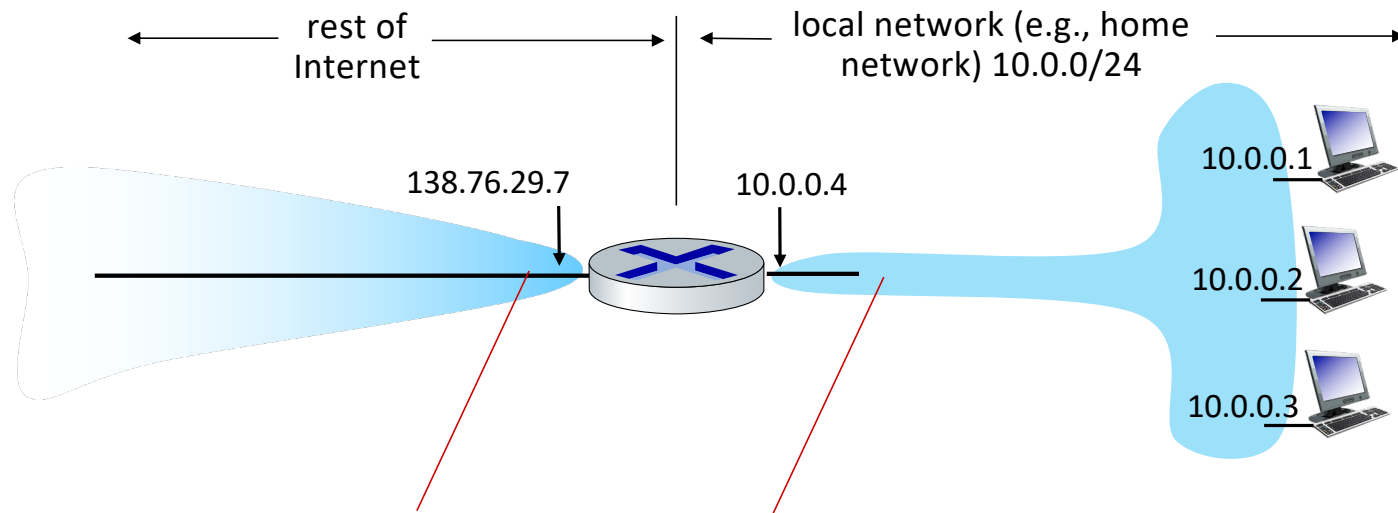
"Who the hell knew how much address space we needed?" Vint Cerf (reflecting on decision to make IPv4 address 32 bits long)

Network layer: “data plane” roadmap

- Network layer: overview
 - data plane
 - control plane
- What’s inside a router
 - input ports, switching, output ports, buffer management, scheduling
- IP: the Internet Protocol
 - datagram format, addressing
 - network address translation, IPv6
- Generalized Forwarding, SDN
 - match+action, OpenFlow: match+action in action
- Middleboxes

NAT: network address translation

NAT: all devices in local network share just **one** IPv4 address as far as outside world is concerned



all datagrams *leaving* local network have *same* source NAT IP address: 138.76.29.7, but *different* source port numbers

datagrams with source or destination in this network have 10.0.0/24 address for source, destination (as usual)

NAT: network address translation

- all devices in local network have 32-bit addresses in a “private” IP address space (10/8, 172.16/12, 192.168/16 prefixes) that can only be used in local network
- advantages:
 - just **one** IP address needed from provider ISP for *all* devices
 - can change addresses of host in local network without notifying outside world
 - can change ISP without changing addresses of devices in local network
 - security: devices inside local net not directly addressable, visible by outside world

NAT: network address translation

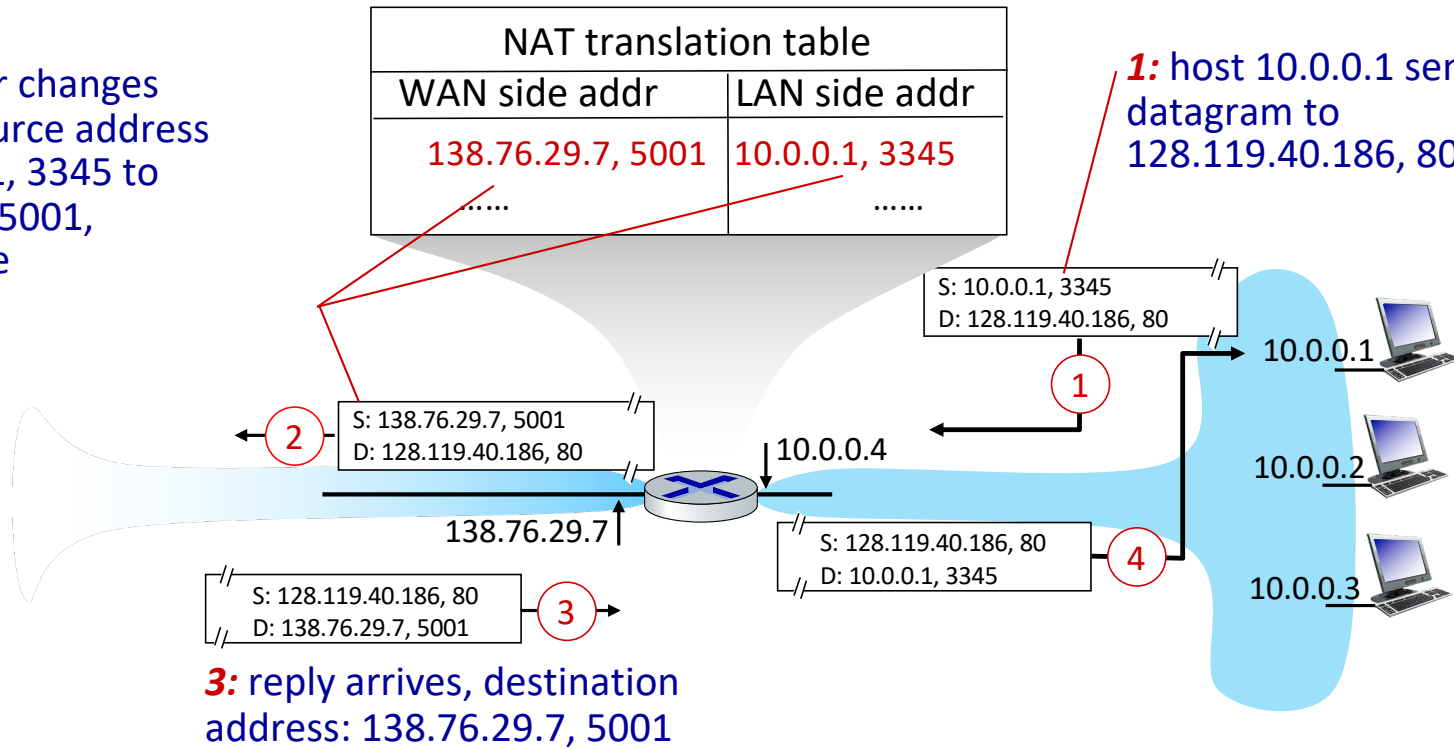
implementation: NAT router must (transparently):

- **outgoing datagrams: replace** (source IP address, port #) of every outgoing datagram to (NAT IP address, new port #)
 - remote clients/servers will respond using (NAT IP address, new port #) as destination address
- **remember (in NAT translation table)** every (source IP address, port #) to (NAT IP address, new port #) translation pair
- **incoming datagrams: replace** (NAT IP address, new port #) in destination fields of every incoming datagram with corresponding (source IP address, port #) stored in NAT table

NAT: network address translation

2: NAT router changes datagram source address from 10.0.0.1, 3345 to 138.76.29.7, 5001, updates table

1: host 10.0.0.1 sends datagram to 128.119.40.186, 80



NAT: network address translation

- NAT has been controversial:
 - routers “should” only process up to layer 3
 - violates end-to-end argument (port # manipulation by network-layer device)
 - address “shortage” should be solved by IPv6
 - NAT traversal: what if client wants to connect to server behind NAT?
- but NAT is here to stay:
 - extensively used in home and institutional nets, 4G/5G cellular nets

Quiz

Suppose a router stores the following IP lookup table:

1010.xxxx.xxxx.xxxx (/4) --> Port 2

10xx.xxxx.xxxx.xxxx (/2) --> Port 1

1010.11xx.xxxx.xxxx (/6) --> Port 3

A packet with destination address 1010.1000.0100.1000 arrives.
Which output port would the router forward the packet to?